

Sarah Henia

Software Engineer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Software engineer who started in full-stack JavaScript (MERN) and now builds production systems in C#.NET, Angular and Python: an enterprise real-time presence library, data pipelines, and AI agents deployed on AWS and Azure. Cares about code that other teams can install, test, trust and maintain.

EXPERIENCE

The SamurAI

Remote, USA

Software Architect & AI Solution Integrator

2026 - Present

- Designed and built **Dawn**, an autonomous **AI agent** for project management, live in production from March 2026: turns recorded meetings into tasks with owner, priority and due date, with **human-in-the-loop** approval before any write to Microsoft Planner.
- Built **multi-model routing** through OpenRouter with automatic fallback; deployed on hardened **AWS EC2** with **GitHub Actions CI/CD** and **observability** in Logfire and Langfuse.
- Designing and building a **multi-agent market-intelligence system** for a Middle East cybersecurity client: bilingual (English/Arabic) **data ingestion**, classification and schema-enforced **LLM extraction** into a **SurrealDB knowledge graph**, with human review before any CRM write.

Product Manager

2025

- Managed the roadmap and backlog of **Dojo**, an AI-powered platform that generates on-demand test labs, turning stakeholder needs into specifications and sprint priorities.

CED Tunisia

Tunis, Tunisia

Software Engineering Intern

Feb - Aug 2026

- Built **Collaboris**, a real-time user-presence library for enterprise web apps, shipped as two installable packages: an **Angular 19** library (npm) and an **ASP.NET Core 8** SDK (NuGet) over **SignalR** and **Redis**. Delivered in three releases under department-head review.
- Designed a five-layer identity cascade that recognises users from the host app's existing session, so nobody signs in twice.
- Built a **layered view resolver**: a scoped, debounced DOM **MutationObserver** that locates each user down to the open dialog, tab or wizard step, so co-presence is exact rather than per-route.
- Detected real user activity by wrapping the browser's **fetch** and **XMLHttpRequest** (write methods only, exclusion list, deduplication window) and multiplexed every tab onto one WebSocket with a **SharedWorker**.
- Built **PresenceAI**, the data and AI layer: **change-data capture** streaming live Redis signals into a Cosmos DB journal (**OLTP to OLAP**, Python/FastAPI), queried in natural language through a governed **Azure OpenAI** assistant over **MCP**.
- Implemented **GDPR compliance by design** (data minimisation, consent, 30-day TTL retention, fail-closed access, cost caps) and covered the AI layer with **unit tests**.
- Deployed the full stack to **Azure** (App Service, SQL Server, Azure Cache for Redis) through **Azure DevOps** build and release pipelines, migrating the API's data layer from PostgreSQL to **SQL Server**.

EDUCATION

Engineering Degree (Diplôme d'Ingénieur), Software Engineering · ESPRIT, Tunis

2023 - 2026

Bachelor's Degree, Business Intelligence · ESSECT, Tunis

2020 - 2023

SKILLS

Backend: C#, ASP.NET Core, SignalR, Python, FastAPI, Node.js, Express, Django, REST, GraphQL, WebSockets, JWT, API gateway

Frontend: TypeScript, JavaScript, Angular, React, Next.js, Tailwind CSS, browser APIs (MutationObserver, SharedWorker), UI/UX, Figma

Data: SQL Server, PostgreSQL, MongoDB, MySQL, Redis, Cosmos DB, SurrealDB; ETL, change-data capture

Cloud & DevOps: Azure (App Service, DevOps pipelines), AWS EC2, Docker, GitHub Actions, CI/CD, Nginx, Linux, npm and NuGet packaging

Testing & AI: pytest, Storybook, ESLint, unit testing, QA; LLM integration, RAG, MCP, AI gateway

CERTIFICATIONS

Cisco CCNA (2024) · AWS Certified Cloud Practitioner (2025) · NVIDIA DLI: Building RAG Agents with LLMs (2025)

LANGUAGES

Arabic (native) · English (C1) · French (B2)